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Fall prevention devices and methods

Abstract

A gate/barrier assembly includes a first post, a second post, a gate, and a barrier. The first post is arranged to be fixedly supported by a semiconductor processing system. The second post extends in parallel with the first post and is arranged to be pivotably supported by the semiconductor processing system. The gate is pivotably supported by the first post and has a closed position where the gate overlaps the second post and an open position wherein the gate is spaced apart from the second post. The barrier is fixedly supported by the second post, has guard and guide positions, is spaced apart from the first post in the guide position, and abuts the first post in the guard position. The gate overlaps the barrier when the gate is in the closed position and the barrier is in the guard position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4150510	12/1978	Harrison	52/64	E06B 11/00
5546703	12/1995	Conway	49/203	E06B 11/00
5592779	12/1996	Conway	N/A	N/A
5855045	12/1998	Miura	N/A	N/A
6193459	12/2000	Rush	N/A	N/A
6212826	12/2000	Kimener	N/A	N/A
6514022	12/2002	Truckor	N/A	N/A
6837296	12/2004	Kimener	N/A	N/A
6976557	12/2004	Becker	182/113	E04G 5/14
7854092	12/2009	Stoffels	49/109	E06B 11/00
8092138	12/2011	Duer	N/A	N/A
8944739	12/2014	Chen	N/A	N/A
9027712	12/2014	Melton	182/36	B65G 69/22
10519014	12/2018	Conway	N/A	N/A
10697236	12/2019	King	N/A	N/A
11761217	12/2022	Curtis	182/113	E04G 5/145
2019/0161980	12/2018	Walker, II	N/A	E04G 1/24
2024/0091989	12/2023	Møller	N/A	E04G 5/141

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201927583	12/2010	CN	N/A
201212147	12/2011	TW	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 63/139,675 filed Jan. 20, 2021, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to fall prevention. More particularly, the present disclosure relates to preventing falls from semiconductor processing systems, such as during the installation and/or removal of modules from semiconductor processing systems.

BACKGROUND OF THE DISCLOSURE

(2) Semiconductor process tools, such as semiconductor process tools used to fabricate semiconductor devices on silicon wafers, commonly employ substrate handling modules and processing modules to manipulate and process silicon substrates. The substrate handling modules generally transfer wafers into the out of the process tool and/or among processing modules of the semiconductor process tool. The processing modules typically form features on the substrates to fabricate the semiconductor device, such as by depositing films onto substrates and/or by etching substrates.

(3) Sometimes, it may be necessary to remove and replace a module in a semiconductor process tool. For example, in some semiconductor processing systems, it may be necessary to remove and replace a module to add new capability to the semiconductor process tool. Alternatively (or additionally), it may be necessary to remove and replace a module for reliability or availability reasons, such as in the unlikely event of a failure. Removal and replacement events may, in some semiconductor process tools, require that technician(s) work from elevated workspaces during the removal and replacement event. Such work may expose the technician(s) to risk of injury in the event of fall from the elevated workspace.

(4) Various devices exist to limit the risk of injury from falls from elevated workspaces. For example, safety railings may be positioned in and around elevated workspace. Safety netting may be positioned around elevated workspaces. And safety tie-offs distributed within the elevated workspace may cooperate with safety harnesses worn by technicians doing work from elevated workspaces to limit risk of injury in the event of fall from the elevated workspace.

(5) Such fall prevention devices and fall prevention methods have generally be acceptable for their intended purpose. However, there remains a need in the art for improved fall prevention devices and methods of preventing falls from elevated workspaces. The present disclosure provides a solution to this need.

SUMMARY OF THE DISCLOSURE

(6) A gate/barrier assembly for a semiconductor processing system is provided. The gate/barrier assembly includes a first post configured to be fixedly supported by a semiconductor processing system; a second post extending in parallel with the first post and configured to be pivotably supported by the semiconductor processing system; a gate pivotably supported by the first post and having a closed position and an open position, the gate overlapping the second post in the closed position, the gate spaced apart from the second post in the open position; and a barrier fixedly supported by the second post and having a guard position and a guide position, the barrier abutting the first post in the guard position, the barrier spaced apart from first post in the guide position. The gate overlaps the barrier when the gate is in the closed position and the barrier is in the guard position.

(7) In certain examples, the gate/barrier assembly may include a biasing member arranged between the gate and the first post. The biasing member may urge the gate toward the closed position to automatically move the gate between the closed position and the open position.

(8) In certain examples, the gate/barrier assembly may have a first post flange. The first post flange may extend about the first post to fixedly support the first post adjacent to a passthrough of the semiconductor processing system.

(9) In certain examples, the gate/barrier assembly may have a second post flange. The second post flange may extend about the second post to pivotably support the second post opposite the first post. The second post may be spaced from the first post by a passthrough of the semiconductor processing system.

(10) In certain examples, the gate/barrier assembly may include a gate midrail. The gate midrail may extend horizontally relative to the first post. The gate midrail may be spaced from the semiconductor processing system by less than 500 millimeters (about 19.7 inches). A gate top rail may extend in parallel with the gate midrail. The gate top rail may be spaced from the gate midrail by less than 500 millimeters (about 19.7 inches).

(11) In certain examples, the gate may include an inboard portion and an outboard portion. The inboard portion may be pivotably supported by the first post. The outboard portion may be pivotably supported by the inboard portion of the gate and therethrough by the first post. The outboard portion of the gate may have an extended position and a folded position, the outboard portion of the gate abutting the inboard portion of the gate in extended position, the outboard portion of the gate overlaying the inboard portion of the gate in the folded position.

(12) In certain examples, the gate/barrier assembly may include a barrier midrail. The barrier midrail may be fixed to the second post. The barrier midrail may extend horizontally from the second post. The barrier midrail may be spaced from the semiconductor processing system by less than 500 millimeters (about 19.7 inches). A barrier top rail may be fixed to the second post. The barrier top rail may extend in parallel with the barrier midrail. The barrier top rail may be spaced from the barrier midrail by less than 500 millimeters (about 19.7 inches).

(13) In certain examples, the gate/barrier assembly may include a first bracket. The first bracket may be fixed to the semiconductor processing system. The bracket may define therein two or more first bracket fastener apertures. The two or more first post fasteners may extend through the two or more first bracket fastener apertures. The two or more first post fasteners may fix the first post to the first bracket.

(14) In certain examples, the gate/barrier assembly may include a second bracket. The second bracket may be fixed to the semiconductor processing system. The second bracket may define therein two or more second bracket fastener apertures. Two or more second post fasteners may extend through the two or more second bracket fastener apertures. The two or more second post fasteners may fix the second post to the second bracket.

(15) In certain examples, the number of second bracket fastener apertures may be greater than the number of second post fasteners to pivotably fix the barrier to the semiconductor processing system for moving the barrier between the guard position and the guide position.

(16) In certain examples, the number of second post threaded apertures may be greater than the number of second post fasteners to pivotably fix the barrier to the semiconductor processing system for moving the barrier between the guard position and the guide position.

(17) In certain examples, the gate/barrier assembly may include a latch member. The latch member may be fixed to one of the barrier and the gate to fix the gate to the barrier when the gate is in the closed position.

(18) In certain examples, the gate may overlap an inboard facing surface of the second post when in the closed position. The barrier may abut a surface of the first post facing the second post when in the closed position.

(19) In certain examples, the gate may be fixed to the barrier when the barrier is in the guard position. The gate may be fixed to the barrier when the barrier is in the guide position.

(20) In certain examples, the gate/barrier assembly may include a hinge. The hinge may connect the gate to the first post. A biasing member may be arranged between the gate and the first post. The biasing member may be configured to urge the gate toward the closed position. A catch mechanism may be operably associated with the hinge and configured to retain the gate in the open position.

- (21) A semiconductor system is provided. The semiconductor processing system includes a wafer handling chamber, a wafer handler, and a gate/barrier assembly as described above. The wafer handling chamber is positioned above a floor and has a workspace and a passthrough that are elevated relative to the floor. The wafer handler is housed within the wafer handling chamber. The first post of the gate/barrier assembly is fixed to the wafer handling chamber, the second post of gate/barrier assembly is pivotably fixed to the wafer handling chamber, and the second post is spaced from the first post by the passthrough. The gate of the gate/barrier assembly spans the passthrough in the closed position and the barrier of the gate/barrier assembly spans the passthrough in the guard position.
- (22) In certain examples, the gate of the gate/barrier assembly may extend inboard from the passthrough in the open position. The barrier of the gate/barrier assembly may extend outboard from the passthrough in the guide position.
- (23) In certain examples, the semiconductor processing system may include a staircase. The staircase may be connected to the wafer handling chamber to provide access to the workspace from the floor. A rail may extend above the wafer handling chamber. The rail may be supported by the wafer handling chamber. The rail may extend above the passthrough between an inboard hoisting location and an outboard hoisting location. A trolley-mounted hoist may be slidably disposed along the rail for transfer of the wafer handler module into and from the workspace.
- (24) A method of removing and replacing a module in a semiconductor processing system is provided. The method includes, at a semiconductor processing system including a gate/barrier assembly as described above, hoisting a wafer handler module from the semiconductor processing system. The gate of the gate/barrier assembly is moved to the open position, the barrier of the gate/barrier assembly is moved to the guide position; and the wafer handler module is transferred through the passthrough of the semiconductor processing system. The gate of the gate/barrier assembly automatically moves to the closed position as the wafer handler module traverses the passthrough.
- (25) In certain examples, the method may include removably fixing the gate to the barrier while the gate is in the closed position and the barrier is in the guide position.
- (26) In certain examples, the method may include moving the gate from the open position by releasing a catch mechanism operably associated with the gate.
- (27) In certain examples, the method may include moving an outboard portion of the gate from an extended position to a folded position. Once in the folded position, a chamber lid may be transferred through space previously occupied by the outboard portion in the extended position.
- (28) This summary is provided to introduce a selection of concepts in a simplified form. These concepts are described in further detail in the detailed description of examples of the disclosure below. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.
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Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) These and other features, aspects, and advantages of the present disclosure herein are described below with reference to the drawings of certain embodiments, which are intended to illustrate and not to limit the present disclosure.
- (2) FIG. 1 is a perspective view of a semiconductor processing system having an elevated workspace and a module passthrough, showing a gate/barrier assembly connected to the semiconductor processing system at the passthrough to prevent falls from the elevated workspace;
- (3) FIG. 2 is a perspective view of a portion of the semiconductor processing system of FIG. 1 including the gate/barrier assembly, showing a gate and a barrier of the gate/barrier assembly

- spanning the passthrough in a closed position and a guard position, respectively;
- (4) FIG. 3 is a perspective view of the gate/barrier assembly of FIG. 1 looking toward the gate/barrier assembly from outboard of the semiconductor processing system, showing the gate pivotably supported by a first post and the barrier fixedly supported by a second post;
- (5) FIG. 4 is a perspective view of the gate/barrier assembly of FIG. 1 looking toward the gate/barrier assembly from inboard of the semiconductor processing system, showing a hinge connecting the gate to the first post and a spring operatively connected to the gate;
- (6) FIG. 5 is a perspective view of a portion of the semiconductor processing system of FIG. 1 including the gate/barrier assembly, showing a wafer handler module being hoisted from the semiconductor processing system while the gate and the barrier span the passthrough in the closed position and the guard position, respectively;
- (7) FIG. 6 is a perspective view of a portion of the semiconductor processing system of FIG. 1 including the gate/barrier assembly, showing the gate in an open position and the barrier in the guard position for movement of the wafer handler module to the passthrough;
- (8) FIG. 7 is a perspective view of a portion of the semiconductor processing system of FIG. 1 including the gate/barrier assembly, showing the barrier in a guide position while the gate remains in the open position in preparation for transfer of the wafer handler module through the passthrough to an outboard hoisting location;
- (9) FIG. 8 is a perspective view of a portion of the semiconductor processing system of FIG. 1 including the gate/barrier assembly, showing the gate automatically moving to the closed position as a technician on the floor adjacent to the passthrough transfers the wafer handler module to the outboard hoisting location;
- (10) FIG. 9 is a perspective view of a portion of the semiconductor processing system of FIG. 1 including the gate/barrier assembly, showing a technician lowering the wafer handler module from the semiconductor processing system from the elevated workspace while the gate remains in the closed position and the barrier remains in the guide position;
- (11) FIGS. 10 and 11 are perspective and plan views of the gate/barrier assembly according to another example the present disclosure, showing a split gate in an extended position and a folded position during removal of a cover overlaying the wafer handling module, respectively; and
- (12) FIG. 12 is a block diagram of a method of removing and replacing a module in a semiconductor processing system, showing operations of the method according to an illustrative and non-limiting example of the method.
- (13) It will be appreciated that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the relative size of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of illustrated embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(14) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an example of a gate/barrier assembly in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character **100**. Other embodiments of gate/barrier assemblies, semiconductor processing systems including gate/barrier assemblies, and methods of removing and replacing modules in semiconductor processing systems in accordance with the present disclosure, or aspects thereof, are provided in FIGS. 2-12, as will be described. The systems and methods described herein may be used for to prevent falls from elevated workspaces during installation and/or replacement of modules in semiconductor processing systems, such as during installation and/or replacement of wafer handler modules in chemical vapor deposition (CVD) and atomic layer deposition (ALD) semiconductor processing systems, though the present disclosure is not limited to wafer handler modules or to any particular type of semiconductor processing system in general.

(15) Referring to FIG. 1, a semiconductor processing system **10** is shown. The semiconductor processing system **10** is arranged on a floor **12**, e.g., in a cleanroom, and includes a wafer handling chamber **14**, a wafer handler module **16** (shown in FIG. 5), a first processing module **18**, and one or more second processing module **20**. The wafer handling chamber **14** is supported above the floor **12**, houses the wafer handler module **16** and includes a gate valve **22**. The gate valve **22** is configured to transfer substrates, e.g., silicon wafers, into and out of the semiconductor processing system **10** via the wafer handling chamber **14**.

(16) The wafer handler module **16** (shown in FIG. 5) is configured to transfer wafers between the gate valve **22**, the first processing module **18**, and/or the one or more second processing module **20**. In certain examples, the wafer handler module **16** may be a first wafer handler module **16** and the semiconductor processing system **10** may include a second wafer handler module **40** housed within the wafer handling chamber **14**. For example, the first wafer handler module **16** may be a front-end wafer handler module and the second wafer handler module **40** may be a back-end wafer handler module. A loadlock module may be arranged between the first wafer handler module **16** and the second wafer handler module **40**.

(17) The first processing module **18** and the one or more second processing module **20** are configured for depositing films and/or etching substrates received from the wafer handler module **16** (shown in FIG. 5). In this respect the first processing module **18** and the one or more second processing module **20** are arranged on the floor **12**, about the wafer handling chamber **14**, and are laterally adjacent to one another. In certain examples, at least one of the first processing module **18** and second processing module **20** may be a CVD processing module. In accordance with certain examples, at least one of the first processing module **18** and the second processing module **20** may be an ALD processing module. In the illustrated example the semiconductor processing system has eight (8) processing modules. This is for illustration purposes and is non-limiting. As will be appreciated by those of skill in the art in view of the present disclosure, semiconductor processing systems having fewer than eight (8) processing modules or more than eight (8) processing modules may also benefit from the present disclosure.

(18) With reference to FIG. 2, as has been explained above, it may sometimes be necessary to install/replace modules from a semiconductor processing system, e.g., the wafer handler module **16** (shown in FIG. 5), such as to upgrade or repair a module of the semiconductor processing system. To facilitate installation/removal of the wafer handler module **16** the semiconductor processing system **10** has a workspace **24** and a passthrough **26**, and includes a staircase **28**, a rail **30**, and a trolley-mounted hoist **32**. The workspace **24** is defined along the top of the wafer handling chamber **14** and is located above the wafer handler module **16**. The workspace **24** extends between an inboard hoisting location **34** and an outboard hoisting location **36**, and traverses the passthrough **26**. It is contemplated that the workspace **24** be further accessible by technicians through the staircase **28**, which is supported by the wafer handling chamber **14** to provide access to the workspace **24** for technicians from the floor **12**.

(19) The passthrough **26** is located within a gap defined between the first processing module **18** (shown in FIG. 1) and the second processing module **20** (shown in FIG. 1), is elevated with respect to the floor **12**, and separates the inboard hoisting location **34** from an outboard hoisting location **36** located above the floor **12**. The elevated workspace **24** extends along an upper surface of the wafer handling chamber **14** to the passthrough **26**, both the elevated workspace **24** and the passthrough **26** being elevated relative to the floor **12**. The rail **30** is supported above the semiconductor processing system **10**, e.g., by frames extending upwards from and supported by the wafer handling chamber **14** and spans an upper surface of the wafer handling chamber **14** between the inboard hoisting location **34** and the outboard hoisting location **36**. The trolley-mounted hoist **32** is slidably disposed along the rail **30**, carries a hoisting mechanism thereon for hoisting and lowering the wafer handler module **16**, and is movable along the rail **30** between the inboard hoisting location **34** and the outboard hoisting location **36**. In certain example, the workspace **24** may be elevated above the

floor **12** by a height sufficient to invoke the safeguard prescribed in SEMI S2-0818, ENVIRONMENTAL, HEALTH, AND SAFETY GUIDELINE FOR SEMICONDUCTOR MANUFACTURING EQUIPMENT, approved for publication on Aug. 17, 2018 (hereafter “SEMI S2-0818”). In accordance with certain examples, the passthrough **26** may be elevated above the floor **12** by about 1.2 meters (about 4 feet). It is contemplated that the trolley-mounted hoist **32** may include a manual hoisting mechanism, e.g., a crank-operated hoisting mechanism.

(20) As has been explained above, work in elevated workspaces may be accompanied by the risk of injury due to falls. For example, technicians working in the elevated workspace **24** may risk injury in the event of fall from the top of the wafer handling chamber **14**. Technicians working on the elevated workspace **24** may also risk injury when installing and/or removing the wafer handler module **16** from the wafer handling chamber **14**. And technicians working on the elevated workspace **24** may further risk injury when transferring the wafer handling module to and from and the semiconductor processing system **10** through the passthrough **26**. To limit (or eliminate) risk of injury due to fall during one or more of these tasks, the semiconductor processing system **10** includes the gate/barrier assembly **100**.

(21) With reference to FIG. **3**, the gate/barrier assembly **100** is configured for placement of at least one (or both) of a barrier **110** (shown in FIG. **3**) and a gate **102** (shown in FIG. **4**) in the passthrough **26** to prevent (or eliminate entirely) risk of fall from the elevated workspace **24** through the passthrough **26**. In this respect the gate/barrier assembly **100** includes the gate **102**, a first post **104**, a hinge **106** (shown in FIG. **4**) and a biasing member **108** (shown in FIG. **4**). As also shown in FIG. **3**, the gate/barrier assembly **100** also includes the barrier **110**, a second post **112** and a latch member **114**. As further shown in FIG. **3**, the gate/barrier assembly **100** further includes a first bracket **116**, a second bracket **118**, a plurality of first post fasteners **120** and a plurality of second post fasteners **122**.

(22) The first post **104** has a first post lower portion **124**, a first post upper portion **126** and a first post flange **128**. The first post lower portion **124**, the first post upper portion **126**, and the first post flange **128** are each arranged along a first post axis **130**. The first post flange **128** extends about the first post **104** and the first post axis **130**, separates the first post lower portion **124** from the first post upper portion **126**, and is configured for seating against the first bracket **116**. The first post lower portion **124** extends upwards from the first post flange **128**, i.e., in a direction axially opposite the first post upper portion **126**, defines therein a plurality of first post threaded apertures **132**, and is configured to seat in the first bracket **116**. The plurality of first post threaded apertures **132** are axially spaced from one another along the first post axis **130** at a common circumferential position about the first post axis **130** for fixing the first post **104** to the first bracket **116**.

(23) The first bracket **116** is configured for fixation to the semiconductor processing system **10** (shown in FIG. **1**). More specifically, the first bracket **116** is configured to fix the first post **104** to the semiconductor processing system **10** at a location below the passthrough **26** (shown in FIG. **2**) such that the first post **104** laterally bounds the passthrough **26**. In this respect the first bracket **116** is connected to the wafer handling chamber **14** below the passthrough **26**, defines therein a first bracket socket **134**, and has a plurality of first bracket fastener apertures **136**. The plurality of first bracket fastener apertures **136** extend through the first bracket **116** to the first bracket socket **134** and are configured to receive therethrough the plurality of first post fasteners **120** for fixation of the first post **104** to the first bracket **116**.

(24) The first post lower portion **124** is slidably received in the first bracket socket **134** such that the first post flange **128** seats against an upper surface of the first bracket **116**. Seating the first post flange **128** against the upper surface of the first bracket **116** positions the first post threaded apertures **132** and the first bracket fastener apertures **136** at a common axial location about the first post axis **130**. The common axial location facilitates assembly of the first post **104** in the first bracket **116** by registering the first bracket fastener apertures **136** with the first post threaded apertures **132**, simplifying the insertion of the plurality of first post fasteners **120** through the

plurality of first bracket fastener apertures **136** and seating within the plurality of first post threaded apertures **132**. Threaded engagement of the plurality of first bracket fasteners **120** fixedly supports the first post **104** at the passthrough **26** (shown in FIG. 1) via the first bracket **116**.

(25) In certain examples, the number of first post fasteners **120** may be equivalent to the number first bracket fastener apertures **136**. In accordance with certain examples, the number of first post fasteners **120** may be equivalent to the number of first post threaded apertures **132**. It is also contemplated that, in certain examples, the number of first post fasteners **120** may be equivalent to both the number of first bracket fastener apertures **136** and the number of first post threaded apertures **132**. As will be appreciated by those of skill in the art in view of the present disclosure, matching the number of fasteners with fastener apertures may error-proof assembly of the first post **104** in the first bracket **116** by limiting the installation to a single orientation about the first post axis **130**. It is contemplated that, in certain examples, the first post lower portion **124** and first bracket socket may have a key portion and corresponding keyway portion to further simplify assembly of the gate/barrier assembly **100** on the semiconductor processing system **10**.

(26) With reference to FIG. 4, the first post upper portion **126** extends axially upwards from the flange **128** along the first post axis **130**. The hinge **106** is connected to the first post upper portion **126** and couples the gate **102** to the first post **104**. In this respect the hinge **106** pivotably fixes the gate **102** to the first post **104** along the first post upper portion **126** such that the gate **102** is pivotable relative to the first post **4** between an open position **138** (shown in FIG. 6) and a closed position **140** (shown in FIG. 2). In the open position **138**, the gate **102** extends inboard from the passthrough **26**, is spaced from the second post **112** and provides limited fall prevention protection. In the closed position **140**, the gate **102** spans the passthrough **26**, overlaps an inboard surface of the second post **112**, and limits (or prevents entirely) risk of fall from the workspace **24** and through the passthrough **26**.

(27) The biasing member **108** is connected between the first post **104** and the gate **102** and is configured to exert a biasing force on the gate **102** urging the gate **102** toward the closed position **140** (shown in FIG. 2). As will be appreciated by those of skill in the art in view of the present disclosure, urging the gate **102** toward the closed position **140** enables the gate **102** to close automatically, limiting (or eliminating) risk of fall through the passthrough **26** by eliminating the need for a technician to mount the wafer handling chamber **14** and manually move the gate **102** to the closed position **140** while located on the elevated workspace **24**. In certain examples, the biasing member **108** may include a spring, such as a coil spring. In accordance with certain examples, the biasing member **108** and the hinge **106** may be integrated as a single assembly, simplifying assembly and/or removal of the gate/barrier assembly **100** from the semiconductor processing system **10** (shown in FIG. 1). A fixation member, accessible from the floor **12** (shown in FIG. 1), may further be operably associated with the gate **102** to fix the gate **102** in the open position **138** (shown in FIG. 6).

(28) The gate **102** may have a gate top rail **142**, gate mid rail **144**, a gate first vertical member **146**, and a gate second vertical member **148**. The gate top rail **142** extends horizontally from the first post **104** and has a length substantially equivalent (or equivalent) to a width of the passthrough **26** (shown in FIG. 2). The gate mid rail **144** similarly extends horizontally from the first post **104**, also has a length substantially equivalent (or equivalent) to a width of the passthrough **26** and is further arranged in parallel with the gate top rail **142**. The gate first vertical member **146** connects the gate top rail **142** to the gate mid rail **144**, is substantially orthogonal (or orthogonal) to both the gate top rail **142** and the gate mid rail **144** and extends in parallel with the first post upper portion **126** of the first post **104**. The gate second vertical member **148** connects the gate top rail **142** to the gate mid rail **144**, is substantially orthogonal (or orthogonal) to the gate top rail **142** and the gate mid rail **144** and extends in parallel with the gate first vertical member **146**, and is further horizontally spaced from the gate first vertical member **146** by both the gate top rail **142** and the gate mid rail **144**. It is contemplated that the hinge **106** may be connected to the gate first vertical member **146** to

pivotably support the gate **102** with the first post **103**. It is also contemplated that, in certain examples, the latch member **114** (shown in FIG. **3**) may seat on (or be configured to removably engage) the gate second vertical member **148**.

(29) In certain examples, the gate midrail **144** may be spaced from the top of the wafer handling chamber **14** (shown in FIG. **1**) by less than 500 millimeters (about 19.7 inches). In accordance with certain examples, the gate top rail **142** may be spaced from the gate midrail **144** by less than 500 millimeters (about 19.7 inches). As will be appreciated by those of skill in the art in view of the present disclosure, spacing the gate midrail **144** less than 500 millimeters (about 19.7 inches) from the top of the wafer handling chamber **14** (shown in FIG. **1**), and the gate top rail **142** from the gate midrail **144** by less than 500 millimeters (about 19.7 inches), renders the semiconductor processing system **10** compliant with SEMI S2-0818 when the gate **102** is in the closed position **140** (shown in FIG. **2**).

(30) With continuing reference to FIG. **3**, the second post **112** is similar to the first post **104** and in this respect has a second post lower portion **150**, a second post upper portion **152** and a second post flange **154**. The second post lower portion **150**, the second post upper portion **152** and the second post flange **154** are each arranged along a second post axis **156**. The second post flange **154** extends about the second post **112** and the second post axis **156** and separates the second post lower portion **150** from the second post upper portion **152**. The second post lower portion **150** extends from the second post flange **154** downward, i.e., in a direction opposite the second post upper portion **152** and defines therein a plurality of second post threaded apertures **158**. The second post upper portion **152** extends axially upwards from the second post flange **154**, along the second post axis **156**, and fixedly supports the barrier **110**.

(31) The barrier **110** has a barrier top rail **168**, a barrier midrail **170**, and a barrier vertical member **172**. The barrier top rail **168** extends horizontally from the second post **112** to the barrier vertical member **172** and has a length substantially equivalent (or equivalent) to a width of the passthrough **26** (shown in FIG. **2**). The barrier midrail **170** extends horizontally from the second post **112** to the barrier vertical member **172**, has a length substantially equivalent (or equivalent) to the width of the passthrough **26**, and further extends in parallel with the barrier top rail **168**. The barrier vertical member **172** connects the barrier top rail **168** to the barrier midrail **170**, is orthogonal relative to both the barrier top rail **168** and the barrier midrail **170** and extends in parallel with the second post **112**. In certain examples, the latch member **114** may seat on (or be configured to removably engage) the barrier top rail **168**. In accordance with certain examples, the latch member **114** may seats on (or is removably engaged) to the barrier at a location along the second post axis **156**. As will be appreciated by those of skill in the art, this allows the gate **102** to be fixedly latched to the barrier **110** irrespective of the position of the barrier **110**, e.g., when the barrier **110** is in a guard position **164** (shown in FIG. **2**) wherein the barrier spans the passthrough **26** and a guide position **166** (shown in FIG. **7**) wherein the barrier **110** extends from the passthrough **26**.

(32) In certain examples, the barrier midrail **170** may be spaced from the top of the wafer handling chamber **14** (shown in FIG. **1**) by less than 500 millimeters (about 19.7 inches). In accordance with certain examples, the barrier top rail **168** may be spaced from the barrier midrail **170** by less than 500 millimeters (about 19.7 inches). As will be appreciated by those of skill in the art in view of the present disclosure, spacing (a) the barrier midrail **170** less than 500 millimeters (about 19.7 inches) from the top of the wafer handling chamber **14** (shown in FIG. **1**), and (b) the barrier top rail **168** by less than 500 millimeters (19.7 inches) from the barrier midrail **170**, renders the semiconductor processing system **10** (shown in FIG. **1**) compliant with SEMI S2-0818 when the gate **102** is in the closed position **140** (shown in FIG. **2**).

(33) The second bracket **118** is configured to pivotably fix the second post **112** to the semiconductor processing system **10** (shown in FIG. **1**). More specifically, the second bracket **118** is configured to fix the second post **112** to the semiconductor processing system **10** at a location below the passthrough **26** (shown in FIG. **1**), laterally bounding the passthrough **26**, and spaced

from the first post **104** by the passthrough **26**. In this respect the second bracket **118** is connected to the wafer handling chamber **14** (shown in FIG. **1**) below the passthrough **26**, defines therein a second bracket socket **160**, and has a plurality of second bracket fastener apertures **162**.

(34) The second post lower portion **150** is slidably received in the second bracket socket **160** such that the second post flange **154** seats on top of the second bracket **118**. Seating the second post flange **154** on the top the second bracket **118** positions the second post threaded apertures **158** and the second bracket fastener apertures **162** on common circumferences about the second post axis **156**. Positioning the second post threaded apertures **158** and the second bracket fastener apertures **162** on common circumference allows the second post **112** (and thereby the barrier **110**) to be pivoted in the second bracket **118** about the second post axis **156** to register different pairs of the second post threaded apertures **158** and the second bracket fastener apertures **162**. For example, the second post **112** may pivoted to locate the either the guard position **164** (shown in FIG. **2**) and the guide position **166** (shown in FIG. **7**), and the second post **112** fastened to the second bracket **118** using the plurality of second post fasteners **122** to fix the barrier **110** in either the guard position **164** or the guide position **166**.

(35) In certain examples, two of the plurality of second post threaded apertures **158** may be circumferentially offset from one another about the second post axis **156** for fixation of the barrier **110** in to the second bracket **118** in both the guard position **164** and the guide position **166**. In accordance with certain examples, two of the plurality of second bracket fastener apertures **162** may be circumferentially offset from one another about the second post axis **156** for fixation of the barrier **110** in to the second bracket **118** in both the guard position **164** (shown in FIG. **2**) and the guide position **166** (shown in FIG. **7**). It is contemplated that, in accordance certain examples, the plurality second bracket threaded apertures **162** may be greater than the plurality of second post fasteners **122** for pivotably fixing the barrier to the semiconductor processing system for moving the barrier between the guard position **164** and the guide position **166**. It is further contemplated that the plurality second post threaded apertures **158** may be greater than the plurality of second post fasteners **122** to pivotably fix the barrier **110** to the semiconductor processing system **10** for movement of the barrier **110** between the guard position **164** and the guide position **166**.

(36) Referring now to FIGS. **5-9**, a method **200** of removing and replacing module in a semiconductor processing system, e.g., the wafer handler module **16**, is shown. As shown in FIG. **5**, the gate **102** and the barrier **110** are first positioned in the passthrough **26** by moving the gate **102** and the barrier **110** to the closed position **140** and the guard position **164**, respectively. Moving the gate **102** to the closed position **140** includes (a) pivoting the gate **102** about the first post **104** from open position **138**, and (b) retaining the gate **102** in the passthrough **26** by overlapping the gate second vertical member **148** against an inboard surface of the second post **112**. Moving the barrier **110** to the guard position **164** includes (c) removing the plurality of second post fasteners **122** from the second post **112**; (d) pivoting the second post **112** (and thereby the barrier **110**) about the second post axis **156**, and (e) fixing the second post **112** in the second bracket **118** by threadedly seating the plurality of second post fasteners **122** in the second post **112** through the plurality of second bracket fastener apertures **162** in the second bracket **118**.

(37) Once the gate is moved to the closed position **140** and the barrier **110** moved to the guard position **164**, the wafer handler module **16** is removed from the semiconductor processing system **10**. Removal of the wafer handler module **16** may be accomplished by (f) sliding the trolley-mounted hoist **32** along the rail **30** to the inboard hoisting location **34**, (g) removing covers from the wafer handling chamber **14** to access and rig the wafer handler module **16** for lifting, and (g) hoisting the wafer handler module **16** from the wafer handling chamber **14**, e.g., by manipulating a crank operably associated with the trolley-mounted hoist **32**. Optionally, the gate **102** may be fixed to the barrier **110** by (h) latching the gate **102** to the barrier **110** with the latch member **114** to prevent movement of the gate **102** from the closed position **140**, e.g., in the event that the gate **102** is bumped during removal and/or movement of the wafer handler module **16**. As will be

appreciated by those of skill in the art in view of the present disclosure, moving the gate **102** to the closed position **140** positions the gate **102** in the passthrough **26**. Positioning the gate **102** in the passthrough allows the gate/barrier assembly **100** to limit (or prevent) risk of fall from the workspace **24** through the passthrough **26**.

(38) As shown in FIG. 6, next, the wafer handler module **16** is readied for transfer through the passthrough **26**. Readyng the wafer handler for transfer through the passthrough **26** includes (a) moving the gate **102** to the open position **138** by pivoting the gate **102** about the first post **104** such that the gate **102** extends inboard from the passthrough **26**, (b) fixing the gate **102** in the open position **138**, and (c) moving the trolley-mounted hoist **32** (and thereby the wafer handler module **16**) from the inboard hoisting location **34** to an inboard intermediate location **38** (shown in FIG. 2) adjacent the passthrough **26** and spaced from the outboard hoisting location by the passthrough **26**. Fixation of the gate **102** in the open position **138** may be accomplished, for example, by (d) manipulating a catch mechanism **174** (shown in FIG. 7) positioned below the biasing member **108** and operably associated with the gate **102**. Optionally, movement of the gate **102** to the open position **138** may be preceded by (e) disengaging the latch member **114** from one of the gate **102** and the barrier **110** to unfix the gate **102** from the barrier **110**. Notably, as the barrier **110** remains the guard position **164** during these operations, the gate/barrier assembly **100** also limits (or eliminates) risk of fall from the workspace **24** through the passthrough **26** during these operations.

(39) As shown in FIG. 7, the barrier **110** is thereafter moved to the guide position **166**. Moving the barrier **110** to the guide position **166** includes (a) unfixing the second post **112** from the semiconductor processing system **10** by removing the plurality of second post fasteners **122** (shown in FIG. 3) from the second post **112** and the second post bracket **118**, (b) pivoting the second post **112** (and thereby the barrier **110**) the second post axis **156** (shown in FIG. 3) such that the barrier **110** extends outboard from the passthrough **26**, and (c) fixing the second post **112** to the semiconductor processing system **10** by seating the plurality of second post fasteners **122** in the second post threaded apertures **158** through the second bracket fastener apertures **162** of the second post bracket **118**. Notably, these operations may be accomplished from the floor **12**.

(40) In the guide position **166** the barrier **110** is spaced from the first post **104** such that the barrier **110** extends from the passthrough **26** in an outboard direction. Notably, as plurality of second post fasteners **122** are seated in the second post bracket **118** at a location below the passthrough **26**, are therefore accessible from a technician located on the floor **12**, the gate/barrier assembly **100** further limits (or eliminates entirely) risk of fall from the workspace **24** through the passthrough **26** during movement of the barrier **110** to the guide position **166**.

(41) As shown in FIG. 8, next, the wafer handler module **16** is transferred through the passthrough **26**. Transfer of the wafer handler module **16** through the passthrough is accomplished by moving the gate **102** from the open position **138** (shown in FIG. 7) to an intermediate position **176** (shown in FIG. 7) located between the open position **138** and the closed position **140** (shown in FIG. 2) by pivoting the gate **102** about the first post **104**. The wafer handler module **16** is then moved the passthrough **26** from the inboard intermediate location **38** adjacent to the passthrough **26** to the outboard hoisting location **36**. Notably, as the wafer handler module **16** moves through the passthrough **26**, the gate **102** moves from the intermediate position **176** to the closed position **140**. It is contemplated that movement of the gate **102** to the closed position **140** occur automatically, i.e., without manipulation by a technician, the gate/barrier assembly **100** thereby limiting (or eliminating) risk without requiring a technician to first position the gate **102** in the closed position **140**.

(42) Optionally, moving the gate **102** from the open position **138** to the intermediate position **176** may include (d) releasing the catch mechanism **174** (shown in FIG. 7), the gate **102** thereby being supported in the intermediate position **176** by the wafer handler module **16**. So supported, the biasing force exerted by the biasing member **108** (shown in FIG. 3) drives the wafer handler module **16** through the passthrough **26** by the engagement of the gate **102** against the wafer handler

module **16**. Alternatively, moving the gate **102** from the open position **138** to the intermediate position **176** may include (e) manipulating the wafer handler module **16** by a technician on the floor **12** to move between the inboard intermediate location **38** adjacent to the passthrough **26** and the outboard hoisting location **36**. Notably, as transfer of the wafer handler module **16** may be accomplished from the floor **12**, the gate/barrier assembly **100** also limits (or eliminates) risk of fall from the workspace **24** through the passthrough **26** during transfer of the wafer handler module **16** through the passthrough **26**.

(43) As shown in FIG. **9**, the wafer handler module **16** is thereafter lowered from the outboard hoisting location **36**. Lowering the wafer handler module **16** from the outboard hoisting location **36** includes (a) mounting the staircase **28** to access the workspace **24**, (b) fixing the gate **102** to the barrier **110** with the latch member **114**, and (c) lowering the wafer handler module **16** by manipulating the crank operably associated with the trolley-mounted hoist **32**. As will be appreciated by those of skill in the art in view of the present disclosure, replacement of the wafer handler module **16** in the semiconductor processing system **10** may be accomplished by reversing the operations shown in FIGS. **5-9** while also limiting (or preventing entirely) risk of fall from the workspace **24** through the passthrough **26**.

(44) With reference to FIGS. **10** and **11**, a gate/barrier assembly **300** is shown. The gate/barrier assembly **300** is similar to the gate/barrier assembly **100** (shown in FIG. **1**) and additionally includes a gate **302** with an inboard portion **304** and an outboard portion **306**. The outboard portion **304** is movable relative to the inboard portion **304** between an extended position **308** (shown in FIG. **10**) and a folded position **310** (shown in FIG. **11**), the outboard portion **306** abutting the inboard portion **304** in the in extended position **308**, the outboard portion **306** overlaying the inboard portion **304** in the folded position **310**. As will be appreciated by those of skill in the art in view of the present disclosure, moving the outboard portion **306** of the gate **302** to the folded position **310** limits intrusion of the gate **302** into the workspace **24** when the gate is in the open position **138** while providing the aforementioned advantages with respect to fall protection. Limiting intrusion of the gate **302** into the workspace **24** in turn allows for removal of relatively large structures from the semiconductor processing system **10** (shown in FIG. **1**), for example chamber lid **42** enclosing the second wafer handler module **40**, eliminating the need to stow such structures in situ, e.g., within the workspace **24**, during removal and/or replacement of the second wafer handler module **40**.

(45) With reference to FIG. **12**, a method **200** of removing and replacing a module in a semiconductor processing system, e.g., the wafer handler module **16** (shown in FIG. **5**), is shown. As shown with box **202**, the method **200** includes moving a gate of a gate/barrier assembly to a closed position, e.g., the gate **102** (shown in FIG. **3**) to the closed position **140** (shown in FIG. **2**). The method **200** also includes moving a barrier of the gate/barrier assembly to a guard position, e.g., the barrier **110** (shown in FIG. **2**) to the guard position **164** (shown in FIG. **5**), as shown with box **204**. It is contemplated that the gate may be fixed to the barrier once in the closed position using a latch member, e.g., the latch member **114** (shown in FIG. **3**), as shown with box **206**. It is also contemplated that gate may be moved to the closed position and/or fixed to the barrier while an elevated workspace of the semiconductor processing system, e.g., the elevated workspace **24** (shown in FIG. **2**), is occupied by a technician, as shown with box **208**.

(46) As shown with box **208**, a module, e.g., the wafer handler module **16** (shown in FIG. **5**), is next hoisted from the semiconductor processing system at an inboard hoisting location, e.g., the inboard hoisting location **34**. Hoisting the module may be accomplished by moving a trolley-mounted hoist to the inboard hoisting location, e.g., the trolley-mounted hoist **32**, removing covers from the wafer handling chamber housing the module, e.g., from the wafer handling chamber **14** (shown in FIG. **1**). Hoisting may be accomplished while by a technician occupying the elevated workspace, such as by manipulating a crank operably connected to the hoist.

(47) In certain examples, hoisting the wafer handler module may include removing a chamber lid,

e.g., the chamber lid **42** (shown in FIG. **11**), from the semiconductor processing system. The gate may then be moved to the open position and folded. Folding may be accomplished by moving an outboard portion of the gate, e.g., from the outboard position **308** (shown in FIG. **11**) to the inboard position **310** (shown in FIG. **11**). Once the gate is folded, the lid may be removed from the semiconductor processing system by shifting the lid through a space occupied by the outboard portion while previously in the extended position.

(48) As shown with box **212**, next, the gate of the gate/barrier assembly is moved from the closed position to an open position, e.g., the open position **138** (shown in FIG. **6**). In certain examples, the moving the gate to the open position may include unfixing the gate from the barrier. In accordance with certain examples, moving the gate to the open position may be accomplished by a technician occupying the elevated workspace. In such examples the barrier provides fall protection to the technician while the technician unfixes the gate from the barrier and/or moves the gate from the closed position to the open position. The module is thereafter transferred from the inboard hoisting location to an inboard intermediate location adjacent to the passthrough, e.g., the inboard intermediate location **38** (shown FIG. **6**), as shown with box **214**. It is contemplated that the transfer be accomplished by a technician occupying the elevated workspace, and that the barrier provide fall protection to the technician during the transfer.

(49) As shown with box **216**, the barrier is next moved from the guard position **164** to a guide position, e.g., the guide position **166** shown in FIG. **7**). It is contemplated that the barrier be moved to the guide position by pivoting a second post of the gate/barrier assembly, e.g., the second post **112** (shown in FIG. **3**), about a second post axis, e.g., the second post axis **156** (shown in FIG. **3**), and while seated in a second bracket, e.g., the second bracket **118** (shown in FIG. **3**). Pivoting second post pivots the barrier about the second post axis. Advantageously, as the barrier is fixed relative to the second post, the barrier may be moved from the guard position to the guide position from the floor, limiting (or eliminating) risk of fall through the passthrough once barrier moves from the guard position. In certain examples, the barrier is moved from the guard position to the guide position while the elevated workspace is unoccupied, as shown with box **218**.

(50) As shown with box **220**, the module is next transferred through the passthrough to an outboard hoisting location, e.g., the outboard hoisting location **36** (shown in FIG. **2**), from the inboard intermediate location. As the module traverses the passthrough the gate moves automatically from the open position to the closed position, as shown with box **222**, e.g., by operation of a biasing member configured to urge the gate toward the closed position, e.g., the biasing member **108** (shown in FIG. **4**). In certain examples, the elevated workspace may be unoccupied as the module traverses the passthrough, e.g., by employment of a lanyard attached to the trolley-mounted hoist and accessible by a technician positioned on the floor adjacent the passthrough. The module is thereafter lowered from the semiconductor processing system at the outboard hoisting location, e.g., by operation of the crank operably connected to the trolley-mounted hoist, as shown with box **222**. In certain examples, lowering may be accomplished by a technician occupying the elevated workspace, fall protection provided by prior automatic movement of the gate to the closed position, as shown with box **224**.

(51) The particular implementations shown and described are illustrative of the present disclosure and its best mode and are not intended to otherwise limit the scope of the aspects and implementations in any way. Indeed, for the sake of brevity, conventional manufacturing, connection, preparation, and other functional aspects of the system may not be described in detail. Furthermore, the connecting lines shown in the various figures are intended to represent exemplary functional relationships and/or physical couplings between the various elements. Many alternative or additional functional relationship or physical connections may be present in the practical system, and/or may be absent in some embodiments.

(52) It is to be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting

sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. Thus, the various acts illustrated may be performed in the sequence illustrated, in other sequences, or omitted in some cases.

(53) The subject matter of the present disclosure includes all novel and nonobvious combinations and subcombinations of the various processes, systems, and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

ELEMENT LISTING

(54) **10** Semiconductor Processing System **12** Floor **14** Wafer handling chamber **16** Wafer Handler Module **18** First Processing Module **20** Second Processing Module **22** Gate Valve **24** Workspace **26** Passthrough **28** Staircase **30** Rail **32** Trolley-Mounted Hoist **34** Inboard Hoisting Location **36** Outboard Hoisting Location **38** Inboard Intermediate Location **40** Second Wafer Handler Module **42** Chamber Lid **100** Gate/Barrier Assembly **102** Gate **104** First Post **106** Hinge **108** Biasing Member **110** Barrier **112** Second Post **114** Latch Element **116** First Bracket **118** Second Bracket **120** Plurality of First Post Fasteners **122** Plurality of Second Post Fasteners **124** First Post Lower Portion **126** First Post Upper Portion **128** First Post Flange **130** First Post Axis **132** First Post Threaded Apertures **134** First Bracket Socket **136** First Bracket Fastener Apertures **138** Open Position (gate) **140** Closed Position (gate) **142** Gate Toprail **144** Gate Midrail **146** Gate First Vertical Member **148** Gate Second Vertical Member **150** Second Post Lower Portion **152** Second Post Upper Portion **154** Second Post Flange **156** Second Post Axis **158** Second Post Threaded Apertures **160** Second Bracket Socket **162** Second Bracket Fastener Apertures **164** Guard Position (Barrier) **166** Guide Position (Barrier) **168** Barrier Toprail **170** Barrier Midrail **172** Barrier Vertical Member **174** Catch Mechanism **176** Intermediate Position **200** Method **202** Box **204** Box **206** Box **208** Box **210** Box **212** Box **214** Box **216** Box **218** Box **220** Box **222** Box **300** Gate/Barrier Assembly **302** Gate **304** Inboard Portion **306** Outboard Portion **308** Extended Position **310** Folded Position

Claims

1. A gate/barrier assembly for a semiconductor processing system, comprising: a first post configured to be fixedly supported by the semiconductor processing system; a second post extending in parallel with the first post and configured to be pivotably supported by the semiconductor processing system; a gate pivotably supported by the first post and having a closed position and an open position, the gate overlapping the second post in the closed position, the gate spaced from the second post in the open position; and a barrier fixedly supported by the second post and having a guard position and a guide position, the barrier abutting the first post in the guard position, the barrier spaced from first post in the guide position; wherein the gate overlaps the barrier when the gate is in the closed position and the barrier is in the guard position.
2. The gate/barrier assembly of claim 1, further comprising a biasing member arranged between the gate and the first post, the biasing member urging the gate toward the closed position to automatically move the gate between the closed position and the open position.
3. The gate/barrier assembly of claim 1, wherein the first post has a first post flange extending about the first post to fixedly support the first post adjacent to a passthrough of the semiconductor processing system.
4. The gate/barrier assembly of claim 1, wherein the second post has a second post flange extending about the second post to pivotably support the second post opposite the first post, the second post spaced from the first post by a passthrough of the semiconductor processing system.
5. The gate/barrier assembly of claim 1, wherein the gate comprises: a gate midrail extending horizontally relative to the first post and spaced from a wafer handling chamber of the semiconductor processing system by less than 500 millimeters (about 19.7 inches); and a gate toprail extending in parallel with the gate midrail, the gate toprail spaced from the gate midrail by

less than 500 millimeters (about 19.7 inches).

6. The gate/barrier assembly of claim 1, wherein the gate comprises: an inboard portion pivotably supported by the first post; an outboard portion pivotably supported by the inboard portion of the gate and therethrough by the first post; wherein the outboard portion of the gate an extended position and a folded position; wherein the outboard portion of the gate abuts the inboard portion of the gate in extended position; and wherein the outboard portion of the gate overlays the inboard portion of the gate in the folded position.

7. The gate/barrier assembly of claim 1, wherein the barrier comprises: a barrier midrail fixed to the second post and extending horizontally from the second post, the barrier midrail from a wafer handling chamber of the semiconductor processing system by less than 500 millimeters (about 19.7 inches); and a barrier top rail fixed to the second post and extending in parallel with the barrier midrail, wherein the barrier top rail and the barrier midrail are spaced from one another by less than 500 millimeters (about 19.7 inches).

8. The gate/barrier assembly of claim 1, further comprising: a first bracket fixed to the semiconductor processing system and defining therein a plurality of first bracket fastener apertures; and a plurality of first post fasteners extending through the plurality of first bracket fastener apertures and fixing the first post to the first bracket.

9. The gate/barrier assembly of claim 1, further comprising: a second bracket fixed to the semiconductor processing system and defining therein a plurality of second bracket fastener apertures; and a plurality of second post fasteners extending through the plurality of second bracket fastener apertures and fixing the second post to the second bracket.

10. The gate/barrier assembly of claim 9, wherein the plurality of second bracket fastener apertures is greater than the plurality of second post fasteners for pivotably fixing the barrier to the semiconductor processing system for moving the barrier between the guard position and the guide position.

11. The gate/barrier assembly of claim 9, wherein the plurality of second post threaded apertures is greater than the plurality of second post fasteners for pivotably fixing the barrier to the semiconductor processing system for moving the barrier between the guard position and the guide position.

12. The gate/barrier assembly of claim 1, further comprising a latch member fixed to one of the barrier and the gate to fix the gate to the barrier when the gate is in the closed position.

13. The gate/barrier assembly of claim 1, wherein the gate overlaps the second post in the closed position.

14. The gate/barrier assembly of claim 1, wherein the gate is fixed to the barrier when the barrier is in the guard position, wherein the gate is fixed to the barrier when the barrier is in the guide position.

15. The gate/barrier assembly of claim 1, further comprising: a hinge connecting the gate to the first post; a biasing member arranged between the gate and the first post, the biasing member configured to urge the gate toward the closed position; and a catch mechanism operably associated with the hinge and configured to retain the gate in the open position.

16. A semiconductor processing system, comprising: a wafer handling chamber positioned above a floor with a workspace and a passthrough, wherein the workspace and the passthrough are elevated relative to the floor; a wafer handler module housed within the wafer handling chamber; and a gate/barrier assembly as recited in claim 1, wherein the first post is fixed to the wafer handling chamber, wherein the second post is pivotably fixed to the wafer handling chamber and spaced from the first post by the passthrough, wherein the gate spans the passthrough in the closed position, and wherein the barrier spans the passthrough in the guard position.

17. The semiconductor processing system of claim 16, wherein the gate extends inboard from the passthrough in the open position, wherein the barrier extends outboard from the passthrough in the guide position.

18. The semiconductor processing system of claim 16, further comprising: a staircase connected to the wafer handling chamber to provide access to the workspace from the floor; a rail extending above the wafer handling chamber and supported by the wafer handling chamber; and a trolley-mounted hoist slidably disposed along the rail, wherein the rail extends above the passthrough between an inboard hoisting location and an outboard hoisting location for transfer of a wafer handler module to and from the workspace.

19. A method of removing and replacing a module in a semiconductor processing system, the method comprising: at a semiconductor processing system having a gate/barrier assembly including a first post fixed to the semiconductor processing system; a second post extending in parallel with the first post and pivotably fixed to the semiconductor processing system; a gate pivotably supported by the first post with a closed position and an open position, the gate spaced from the second post in the open position, the gate overlapping the second post in the closed position; and a barrier fixedly supported by the second post with a guide position and a guard position, the barrier spaced from the first post in the guide position, the barrier abutting the first post in the guard position, and the barrier overlapping the barrier when the gate is in the closed position and the barrier is in the guard position; hoisting a wafer handler module from the semiconductor processing system; moving the gate to the open position; moving the barrier to the guide position; and transferring the wafer handler module through a passthrough of the semiconductor processing system, wherein the gate automatically moves to the closed position as the wafer handler module traverses the passthrough.

20. The method of claim 19, further comprising: fixing the gate to the barrier while the gate is in the closed position and the barrier is in the guide position; and moving the gate from the open position by releasing a catch mechanism operably associated with the gate.
